

19. Dimensions and Column Names of World Alcohol Consumption Dataset

Aim

To write a Pandas program to display the **dimensions/shape** of the World Alcohol Consumption dataset and extract its column names.

Dataset

A simplified version of the **World Alcohol Consumption** dataset is shown below:

Country	Beer_Servings	Spirit_Servings	Wine_Servings	Total_Litres
Afghanistan	0	0	0	0.0
Albania	89	132	54	4.9
Algeria	25	0	14	0.7
Andorra	245	138	312	12.4
Angola	217	57	45	5.9

Python code:

```
1 import pandas as pd
2 # Read the dataset
3 df = pd.read_excel(r"C:\Users\DELL\Documents\alcohol.xlsx")
4 # Display the shape
5 print("Shape of the dataset:")
6 print(df.shape)
7 # Display number of rows and columns
8 print("\nNumber of rows:", df.shape[0])
9 print("Number of columns:", df.shape[1])
10 # Extract column names
11 print("\nColumn Names:")
12 print(df.columns.tolist())
```

Output:

```
Number of rows: 5
Number of columns: 5
```

```
Column Names:
['Country', 'Beer_Servings', 'Spirit_Servings', 'Wine_Servings', 'Total_Litres']
```

Result:

The dimensions and column names of the World Alcohol Consumption dataset were successfully displayed.

20. Find the Index of a Given Substring in a DataFrame Column

Aim:

To write a Pandas program to find the index/position of a given substring in a DataFrame column.

Index	Name	City
0	Anu	Chennai
1	Bala	Madurai
2	Cathy	Coimbatore
3	David	Chennai
4	Esha	Bangalore

Python code:

```

1 import pandas as pd
2 data = {
3     "Name": ["Anu", "Bala", "Cathy", "David", "Esha"],
4     "City": ["Chennai", "Madurai", "Coimbatore", "Chennai", "Bangalore"]
5 }
6 df = pd.DataFrame(data)
7 print("Original DataFrame:")
8 print(df)
9 # Given substring
10 substring = "nai"
11 # Find the index of substring in City column
12 result = df["City"].str.find(substring)
13 print("\nIndex of substring:", substring)
14 print(result)

```

Output

```

Index of substring: nai
0      4
1     -1
2     -1
3      4

2     -1
3      4
4     -1
Name: City, dtype: int64

```

Result:

The index position of the specified substring was successfully found using the Pandas `str.find()` function.